# Heegner points and Kolyvagin's theorem on modular elliptic curves

Graduate Student Seminar, organized by H. Darmon (McGill University)

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Let $K$ be an imaginary quadratic field of discriminant $-D$, where $D \neq 3$ and $4$, and let $N$ be an integer satisfying the Heegner hypothesis in $K/\mathbb{Q}$.

Let $E/\mathbb{Q}$ be an elliptic curve of conductor $N$, and $\varphi : X_0(N) \longrightarrow E$ be a modular parametrization.

**Theorem (Kolyvagin)**

Suppose that $y_K$ has infinite order in $E(K)$. Then:

1. $E(K)$ has rank 1.
2. $Ш(E/K)$ is finite and has order dividing $t_{E/K}(I_K)^2$, where $t_{E/K}$ is an integer whose prime factors depend only on $E$ and $I_K = [E(K) : \mathbb{Z}y_K]$.

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**Theorem (Kolyvagin; weak form)**

Let $p$ be an odd prime such that

$$\mathrm{Gal}(\mathbb{Q}(E_p)/\mathbb{Q}) \cong \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}),$$

and assume that $p$ does not divide $y_K$ in $E(K)$. Then:

1. $E(K)$ has rank 1.
2. $Ш(E/K)_p$ is trivial.

Key proposition: Let $p$ be an odd prime such that $\mathrm{Gal}(\mathbb{Q}(E_p)/\mathbb{Q}) \cong \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$, and assume that $p$ does not divide $y_K$ in $E(K)$. Then:

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Heegner hypothesis: all prime factors of $N$ are split in $K/\mathbb{Q}$.

$\longrightarrow \exists\, \mathcal{N} \leq \mathcal{O}_K$ such that

$$\mathcal{O}_K/\mathcal{N} \cong \mathbb{Z}/N\mathbb{Z}.$$

$\longrightarrow \exists\, \varphi : \mathbb{C}/\mathcal{O}_K \longrightarrow \mathbb{C}/\mathcal{N}^{-1}$ cyclic of order $N$.

**Definition (Heegner point)**

The point $x_1$ defined by

$$x_1 = (\mathbb{C}/\mathcal{O}_K, \mathcal{N}^{-1}/\mathcal{O}_K) \in X_0(N),$$

is called **"Heegner point"** (of conductor 1).

$\longrightarrow$ $x_1$ is rational over $H_K$, the Hilbert class field of $K$.

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Let $n \geq 1$ be an integer such that $n \wedge N = 1$, and $\mathcal{O}_n = \mathbb{Z} + n\mathcal{O}_K$.

Set $\mathcal{N}_n := \mathcal{N} \cap \mathcal{O}_n$. The point

$$x_n = (\mathbb{C}/\mathcal{O}_n, \mathcal{N}_n^{-1}/\mathcal{O}_n) \in X_0(N)(K_n)$$

is called **"Heegner point"** (of conductor $n$).

**Proposition**

Let $\sigma \in \mathrm{Gal}(K_n/K)$. We have:

$$\sigma \cdot (\mathbb{C}/\mathcal{O}_n, \mathcal{N}_n^{-1}/\mathcal{O}_n) = (\mathbb{C}/\alpha_\sigma^{-1}, \mathcal{N}_n^{-1}\alpha_\sigma^{-1}/\mathcal{O}_n)\,,$$

where $\alpha_\sigma$ denotes the class corresponding to $\sigma$ in $\mathrm{Pic}(\mathcal{O}_n)$ via the Artin map.

**Proposition**

Let $n = m\ell$, where $\ell \nmid m$ is prime. Then:

$$\mathrm{Tr}_\ell(x_n) = T_\ell(x_m).$$

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The points $y_n = \varphi(x_n) \in E(K_n)$ are called **"Heegner points"** on $E$. From now on, $n$ is square-free and no prime divisor $\ell$ of $n$ divides $pND$.

Let $\mathrm{Frob}(\ell)$ be the conjugacy class in $\mathrm{Gal}(K(E_p)/\mathbb{Q})$ of each Frobenius of $\mathrm{Gal}(\mathbb{F}_\gamma/\mathbb{F}_\ell)$.

**Definition (Kolyvagin prime)**

A prime $\ell$ as above satisfying the relation

$$\mathrm{Frob}(\ell) = \mathrm{Frob}(\infty)$$

is called a **"Kolyvagin prime"**.

$\longrightarrow$ Let $\lambda$ be the unique prime of $K$ lying above $\ell$.

**Proposition**

If $\ell$ is a Kolyvagin prime, then:

$$a_\ell \equiv \ell + 1 \equiv 0 \pmod{p}.$$

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**Proposition**

If $n = m\ell$, where $\ell$ is a Kolyvagin prime, then:

1. $\mathrm{Tr}_\ell(y_n) = a_\ell y_m \in E(K_m)$;
2. Every prime $\lambda_n$ lying above $\ell$ in $K_n$ lies above a unique prime $\lambda_m$ of $K_m$. Moreover,

$$y_n \equiv \mathrm{Frob}(\lambda_m)(y_m) \pmod{\lambda_n}.$$

Suppose $n = m\ell$ is square-free and $\ell$ is a Kolyvagin prime.

Fix a generator $\sigma_\ell$ of $G_\ell = \mathrm{Gal}(K_n/K_m) \leq G_n = \mathrm{Gal}(K_n/K_1)$. Consider the equation:

$$(\sigma_\ell - 1)X = \ell + 1 - \mathrm{Tr}_\ell.$$

Let $D_n$ be the product of all solutions $D_\ell$. Kolyvagin uses the solution

$$D_\ell^0 = \sum_{i=1}^{\ell} i.\sigma_\ell^i = -\sum_{i=1}^{\ell+1} \frac{\sigma_\ell^i - 1}{\sigma_\ell - 1}.$$

$\longrightarrow$ By construction, $D_n(y_n)$ lies in $E(K_n)$, and the class $[D_n(y_n)] \in E(K_n)/pE(K_n)$ satisfies the relation:

$$\sigma([D_n(y_n)]) = [D_n(y_n)], \quad \forall \sigma \in G_n.$$

Next goal: Construct a point in $E(K_n)/pE(K_n)$ invariant under $\mathcal{G}_n := \mathrm{Gal}(K_n/K)$.

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Here we use the Galois group hypothesis for the first time.

Let $S$ be a set of coset representatives for $G_n$ in $\mathcal{G}_n$. Define $P_n = \sum_{\sigma \in S} \sigma(D_n(y_n)) \in E(K_n)$.

**Lemma**

The curve $E$ has no $p$-torsion rational over $K_n$.

**Corollary**

There is an isomorphism $H^1(K, E_p) \xrightarrow{\sim} H^1(K_n, E_p)^{\mathcal{G}_n}$.

Hochshild-Serre spectral sequence:

$$0 \to H^1(\mathcal{G}_n, H^0(G_{K_n}, E_p)) \overset{\text{Inf}}{\to} H^1(K, E_p) \overset{\text{Res}_1}{\to} H^1(K_n, E_p))^{\mathcal{G}_n} \overset{\text{Trans}}{\to} H^2(\mathcal{G}_n, H^0(G_{K_n}, E_p)).$$

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 & & & & & & 0 & & \\
 & & & & & & \downarrow & & \\
 & & & & & & H^1(K_n/K, E)_p & & \\
 & & & & & & \downarrow{\scriptstyle \mathbf{Inf}} & & \\
0 & \longrightarrow & E(K)/pE(K) & \xrightarrow{\delta} & H^1(K, E_p) & \xrightarrow{\gamma} & H^1(K, E)_p & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle \mathbf{Res_1}} \wr & & \downarrow{\scriptstyle \mathbf{Res_2}} & & \\
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**Definition (cohomology classes $c(n)$)**

Define $c(n) \in H^1(K, E_p)$ as the unique class satisfying

$$\mathrm{Res}_1(c(n)) = \delta_n([P_n]).$$

Let $d(n) = \gamma(c(n))$ and let $\widetilde{d(n)}$ be defined by $\mathrm{Inf}(\widetilde{d(n)}) = d(n)$.

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$\longrightarrow$ $c(n)$ is trivial iff $P_n \in pE(K_n)$.

$\longrightarrow$ $d(n)$ and $\widetilde{d(n)}$ are trivial iff $P_n \in pE(K_n) + E(K)$.

**Fact:** The complex conjugation $\tau$ acts on $y_n$ by:

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0 & \longrightarrow & \prod_v E(K_v)/pE(K_v) & \longrightarrow & \prod_v H^1(K_v, E_p) & \longrightarrow & \prod_v H^1(K_v, E)_p & \longrightarrow & 0
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$$\text{Sel}_p(E/K) = \ker\left(H^1(K, E_p) \xrightarrow{g\circ\gamma} \prod_v H^1(K_v, E)_p\right)$$

$\longrightarrow c(n) \in \text{Sel}_p(E/K) \iff d(n)_v = 0$ for every place $v$ of $K$.

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# Local Tate duality

The Weil pairing induces a cup-product pairing in Galois cohomology:

$$H^1(K_\lambda, E_p) \times H^1(K_\lambda, E_p) \longrightarrow \mathbb{Z}/p\mathbb{Z}.$$

**Local Tate duality:** The pairing $\langle \cdot, \cdot \rangle : H^1(K_\lambda, E_p) \times H^1(K_\lambda, E_p) \longrightarrow \mathbb{Z}/p\mathbb{Z}$ is alternating and non-degenerate.

**Proposition**

The pairing $\langle \cdot, \cdot \rangle$ induces a non-degenerate pairing of $\mathbb{Z}/p\mathbb{Z}$-vector spaces

$$\langle \cdot, \cdot \rangle : E(K_\lambda)/pE(K_\lambda) \times H^1(K_\lambda, E)_p \longrightarrow \mathbb{Z}/p\mathbb{Z}.$$

**Theorem**

Assume that the class $d(n) \in H^1(K, E)_p^{\pm}$ is locally trivial for all places $v \neq \lambda$ of $K$ and $d(n)_\lambda$ is non-trivial in $H^1(K_\lambda, E)_p^{\pm}$. Then, for any class $s \in \mathrm{Sel}_p(E/K)^{\pm} \subset H^1(K, E_p)^{\pm}$,

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# Proof of Kolyvagin's theorem (weak form)

From now on, assume that $D$ is prime to $pN$ and $\mathrm{Gal}(\mathbb{Q}(E_p)/\mathbb{Q}) \cong \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$.

Let $L = K(E_p)$ and set $\mathcal{G} := \mathrm{Gal}(L/K) \cong \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$.

**Fact (Serre):** For all $n \geq 0$, $H^n(\mathcal{G}, E_p) = 0$. Moreover, the restriction induces an isomorphism $H^1(K, E_p) \xrightarrow{\sim} H^1(L, E_p)^{\mathcal{G}}$.

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**Proposition**

The induced pairing $[\cdot,\cdot] : S \times \mathrm{Gal}(L_S/L) \longrightarrow E_p(L)$ is non-degenerate.

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Next goal: Apply these results to $S = \mathrm{Sel}_p(E/K)$.

# Proof of Kolyvagin's theorem (weak form)

From now on, assume that $D$ is prime to $pN$ and $\mathrm{Gal}(\mathbb{Q}(E_p)/\mathbb{Q}) \cong \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$.

Let $L = K(E_p)$ and set $\mathcal{G} := \mathrm{Gal}(L/K) \cong \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$.

**Fact (Serre):** For all $n \geq 0$, $H^n(\mathcal{G}, E_p) = 0$. Moreover, the restriction induces an isomorphism $H^1(K, E_p) \xrightarrow{\sim} H^1(L, E_p)^{\mathcal{G}}$.

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$$[\sigma(s), \sigma(\rho)] = [s, \sigma(\rho)] = \sigma([(s, \rho]),$$

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Let $S$ be a finite subgroup of $H^1(K, E_p)$, and $L_S$ be the fixed field of $\mathrm{Gal}_S(\overline{\mathbb{Q}}/L)$ defined by $\mathrm{Gal}_S(\overline{\mathbb{Q}}/L) = \left\{\rho \in \mathrm{Gal}(\overline{\mathbb{Q}}/L) : [s, \rho] = 0,\ \forall s \in S\right\}$.

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# Proof of Kolyvagin's theorem (weak form)

Hypothesis: $y_K$ is not divisible by $p$ in $E(K)$.

$\longrightarrow$ Let $\delta(y_K)$ be its non-zero image in $\mathrm{Sel}_p(E/K)$.

Let $H$ be the Galois group of $L_S/L$ and $I$ be the subgroup of $H$ fixing the subfield $L\left(\frac{1}{p}y_K\right) = L_{\langle\delta(y_K)\rangle}$ of $L_S$.

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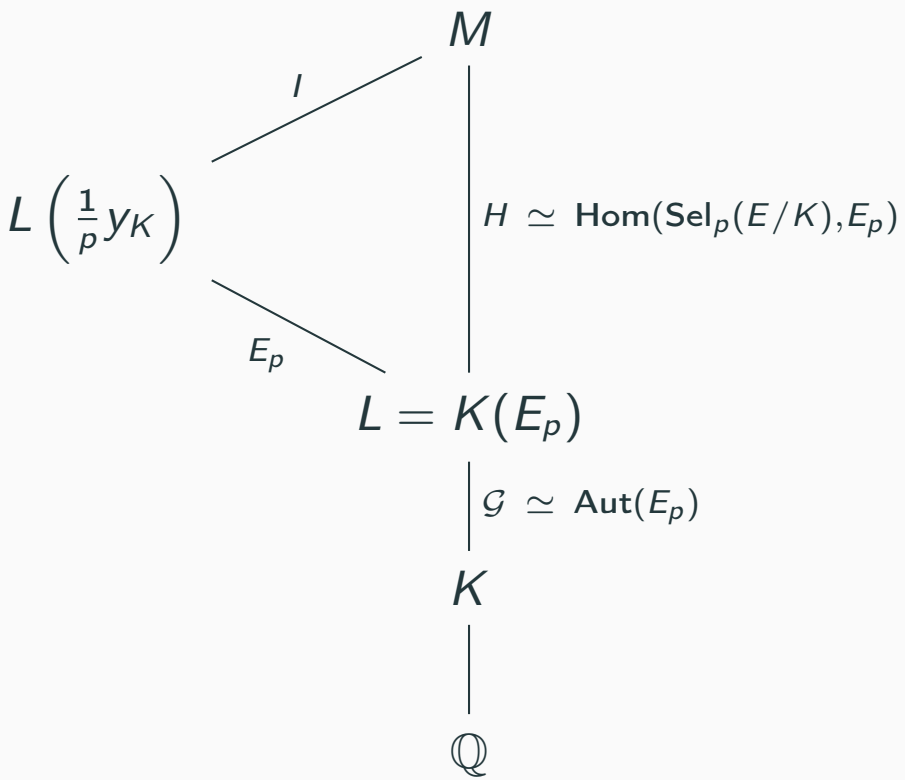
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Let $\tau$ be a fixed complex conjugation in $\mathrm{Gal}(M/\mathbb{Q})$, and $H^+$ and $I^+$ denote the $+1$-eigenspaces for (acting by conjugation) on $H$ and $I$.

$\longrightarrow H^+ = \left\{(\tau h)^2 : h \in H\right\}$, $I^+ = \left\{(\tau i)^2 : i \in I\right\}$ and $H^+/I^+ \cong \mathbb{Z}/p\mathbb{Z}$.

**Proposition**

If $s \in \mathrm{Sel}(E/K)_p^{\pm}$, then the following are equivalent:

1. $[s, \rho] = 0, \ \forall \rho \in H$ ;
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**Proposition**

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Recall: The Heegner point $y_K = P_1$ lies in the $\epsilon$-eigenspace for complex conjugation on $E(K)/pE(K)$.

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Assume that $y_K$ is not divisible by $p$ in $E(K)$. Let $\ell$ be a rational prime which is unramified in $M/\mathbb{Q}$ and has a prime $\lambda_M$ lying above it, and whose Frobenius element is equal to $\tau h$ in $\mathrm{Gal}(M/\mathbb{Q})$, with $h \in H$. The following are all equivalent:

1. $c(\ell) \in \mathrm{Sel}(E/K)_p \subset H^1(K, E_p)$ ;
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$\mathrm{Sel}(E/K)_p^{\epsilon} \cong \mathbb{Z}/p\mathbb{Z}.\delta(y_K)$.

**Consequence (Kolyvagin + Gross-Zagier):**

**Kolyvagin:** If $y_K$ has infinite order in $E(K)$, then $\mathrm{rank}(E/K) = 1$.

**Gross-Zagier:** $L'(E/K, 1) = \frac{\iint_{E(\mathbb{C})} \omega \wedge \overline{i\omega}}{\sqrt{D}} \hat{h}(y_K)$.

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